



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

December 02, 2022

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 9791
(EIGHTEENTH REVISION)

EXPIRATION DATE: 2026-11-30

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: PST Cylinders, LLC
Cudahy, Wisconsin
2. PURPOSE AND LIMITATION:
 - a. This special permit authorizes the manufacture, mark, sale and use of a non-DOT specification cylinder conforming in part with the DOT-3AA specification, for use in the transportation of certain nonflammable, non-liquefied compressed gases transportation in commerce. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. In accordance with 49 CFR 107.107(a), party status may not be granted to a manufacturing permit. These packagings may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171- 180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR §§ 173.301(a)(1), 173.301(a)(2) and 173.302(a)(1) in that non-DOT specification cylinders are not authorized except as prescribed herein.
5. BASIS: This special permit is based on the application of PST Cylinders, LLC dated December 1, 2022, in accordance with § 107.109.

Tracking Number: 2022124059

6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Material Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Nonflammable, non-liquefied gases authorized for DOT 3AA specification cylinders	2.2	As appropriate	N/A

7. SAFETY CONTROL MEASURES:

- a. PACKAGING: Packaging prescribed is a high strength, non-DOT specification steel cylinder, conforming with PST's drawing 1280201-4 dated February 1987 on file with the Office of Hazardous Materials Safety Approvals and Permits Division (OHMS) and DOT specification 3AA (§§ 178.35 and 178.37), except as follows:

§ 178.35(c) *Duties of Inspector.*

* * *

(Add) (5) Verify that the design qualification tests prescribed in § 178.37(d)(3) have been performed with satisfactory results.

(Add) (6) Lot definition. In this special permit, a “lot” means a group of cylinders successively produced and having the same:

- (i) Size and configuration;
- (ii) Specified material of construction;
- (iii) Process of manufacture and heat treatment;
- (iv) Equipment of manufacture and heat treatment;
- (v) Conditions of time, temperature and atmosphere during heat treatment.

In no case may the lot size exceed 200 cylinders, but any cylinder processed for use in the required destructive testing need not be counted as being one of the 200.

§ 178.35(e) *Pressure relief devices and protection for valves and pressure relief devices.*

(Add) Pressure relief devices must be in compliance with § 173.302a(b)(1).

§ 178.35(f) *Marking.*

Applies except that “DOT-SP 9791” must replace “DOT-3A” followed by the service pressure. Test pressure “TP 5250” must be marked following or near the service pressure. The letters “TP” preceding the test pressure is optional.

§ 178.35(g) *Inspector’s report.*

The inspector’s report must be revised to accommodate the above changes in requirements.

§ 178.37(a) *Type, size and service pressure.*

(1) Seamless cylinder with 6.875 inches nominal inside diameter, 0.179 inch minimum wall thickness, 50 pounds maximum water capacity, a maximum service pressure of 3,500 psig, and a test pressure of 5,250 psig.

(2) Does not apply.

§ 178.37(b) *Authorized steel.*

Electric furnace or equivalent steel of uniform quality is authorized. The steel analysis must conform with the following:

Chemical Composition in Weight Percent				
Element	Ladle Analysis	Check Analysis Tolerance		
	Gr. 1	Gr. 2	Under	Over
Carbon	0.31/0.35	0.28/0.33	0.01	0.02
Manganese	0.60/0.90	0.70/0.90	0.03	0.03
Phosphorus	0.025 max.	0.015 max.	-	0.01

Chemical Composition in Weight Percent				
Element	Ladle Analysis	Check Analysis Tolerance		
Sulfur	0.010 max.	0.010 max.	-	0.01
Silicon	0.15/0.35	0.15/0.35	0.02	0.03
Chromium	0.80/1.10	0.80/1.10	0.03	0.03
Molybdenum	0.15/0.25	0.15/0.25	0.01	0.01
Vanadium	0.07/0.10	0.02 max.	0.01	0.01
Aluminum	0.01/0.05	0.02/0.06	-	0.00
Copper	0.20 max	0.20	-	0.00
Sulphur plus-phosphorus	-	0.020 max	-	-

Note 1: Steel shall be treated with calcium to provide the following J-K microcleanliness rating per ASTM Standard E- 45, Method D for Gr. 1 and Method A for Gr.2.

A (Sulfides)		B (Alumina)		C (Silicates)		D (Oxides)	
Thin	Heavy	Thin	Heavy	Thin	Heavy	Thin	Heavy
2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

Certificate from the material manufacturer must certify that the material was calcium treated and must include in such certification the J-K microcleanliness rating for each heat of steel.

Note 2: Steel shall be aluminum killed and made by a fine grained de-oxidation practice.

§ 178.37(c) *Identification of material.*

Materials must be identified by any suitable method. Steel stamping of heat identification must not be made in any area that will eventually become the sidewall of

the cylinder. Depth of stamping shall not encroach upon the minimum prescribed wall thickness of the cylinder.

§ 178.37(d) *Manufacture.*

(1) Cylinder shells must be of seamless construction manufactured by the deep drawing method with integrally formed heads and bottoms; dirt and scale to be removed as necessary to afford proper inspection; no fissure or other defect acceptable that is likely to weaken the finished cylinder appreciably. The general surface finish shall not exceed a roughness of 250 r.m.s., and individual irregularities such as draw marks, scratches, pits, etc. should be held to a minimum. If the cylinder is not originally free of such defects or does not meet the finish requirements, the surface may be machined or otherwise treated to eliminate these defects. Metal removal for any purpose other than removal of isolated defects and threading must be done prior to the hydrostatic test. The thickness of the treated areas must be measured and must not be less than the minimum prescribed thickness. Cylinder end contour must be hemispherical or ellipsoidal (axis ratio of 2:1) with concave side to pressure.

(2) Shape and thickness of the cylinder bottom and sidewall adjacent to the bottom must be such that failure during the cycle pressure test occurs in the sidewall of the cylinder. The thickness of the cylinder bottom must be no less than the cylinder sidewall thickness.

(3) Design qualifications: The design authorized herein and any significant change to this design must be qualified for production by performing the test specified below:

(i) Burst Test: Three cylinders must be hydrostatically burst without evidence of fragmentation. The rate of pressurization must not exceed 200 psi per second. Cylinders subjected to the burst test must withstand a pressure of at least 2.25 times the service pressure without failure. Failure must initiate in the sidewall in a longitudinal direction, and the cylinder must remain in one piece.

(ii) Flattening Test: Three cylinders must be flattened to eight times the wall thickness without cracking.

(iii) Cycle Test: Three cylinders must be cycle tested to destruction to an upper cyclic pressure of 1.5 times service pressure. The successive hydrostatic pressurizations from the lower cyclic pressure to the upper cyclic pressure must not exceed a rate of ten cycles per minute. Adequate recording instrumentation must be provided if equipment is to be left unattended for any period of time. Lower

cyclic pressure must not exceed 10 percent of the upper cyclic pressure. Cylinders must withstand at least 10,000 cyclic pressurizations without distortion or failure. The failure must occur in the sidewall and the failure mode must be leak before burst (LBB). At least one cylinder must be cycled using water as the pressurizing medium.

(iv) Flawed Burst Test: One cylinder must be cycle tested to destruction at a pressure of 1.25 times the service pressure. This test must be performed after three flaws (slots) are machined into the upper sidewall of the cylinder. The flaws must have a minimum length of 6t and be located at 120° intervals. The flaws must be introduced into the cylinder by a means that will not affect the mechanical or metallurgical properties of the cylinder. The failure mode must be LBB. Examination of the failed cylinder must show evidence of fatigue crack propagation prior to leakage.

In this special permit, “significant change” means a 10 percent or greater change in cylinder wall thickness, service pressure or diameter; a 30 percent or greater change in water capacity of base thickness; any change in material; over a 100 percent increase in size of openings; or any change in the number openings.

§ 178.37(e) *Welding or brazing.*

Welding or brazing for any purpose is prohibited.

§ 178.37(f) *Wall thickness.*

(1) Does not apply.

(2) The minimum wall thickness must be such that the wall stress at the minimum specified test pressure does not exceed 67 percent of the minimum tensile strength of the steel as determined in §§ 178.37-16 and 178.37-17 of this special permit. A wall stress of more than 90,500 psi is not permitted and in no case may the wall thickness be less than 0.179 inch.

(3) Calculation must be made by the formula:

$$S = P(1.3D^2 + 0.4d^2)/(D^2 - d^2)$$

where:

S = wall stress in pounds per square inch;

P = minimum test pressure of 3/2 of service pressure; D = outside diameter in inches;

d = inside diameter in inches.

§ 178.37(g) *Heat treatment.*

* * *

(1) Each cylinder must be heated and held above the upper critical temperature (Ac_3) for at least one hour per inch of thickness based on the maximum thickness of the cylinder and then quenched in a suitable liquid medium having a cooling rate not in excess of 80 percent of water. The steel temperature on quenching must be above the Ac_3 temperature but not higher than 1700 °F.

(2) After quenching, each cylinder must be reheated to a temperature below the transformation range but not less than 1000 °F, and must be held at this temperature for at least one hour per inch of thickness based on the maximum thickness of the cylinder. Each cylinder must then be air cooled.

§ 178.37(h) *Openings in cylinders and connections (valves, fuse plugs, etc.) for those openings.*

(1) Threads required, to be clean cut, even, without checks, and to gauge. Openings are permitted in the top head and centerline of the cylinder only.

(2) All openings must be circular and threaded. Straight threads must be used and must conform with the following:

(i) National Gas Straight Thread (NGS) type must conform with the requirements of Federal Standard H-28 (1978), Sections 7 and 9.

(ii) Unified Thread (UN) type must conform with the requirements of Federal Standard H-28 (1978), Section 2.

(iii) Controlled Radius Root Thread (UNJ) type must conform with the requirements of Federal Standard H-28 (1978), Section 4.

(iv) Other straight thread types conforming with other recognized standards may be used provided that the requirements of paragraph (3) below are met.

- (3) All straight threads must have at least 6 engaged threads, a tight fit, and a factor of safety in shear of at least 10 at the test pressure of the cylinder. Shear stress must be calculated by using the appropriate thread shear area in accordance with Federal Standard H-28 (1978), Appendix A5, Section 3. Gaskets are required to prevent leakage.

§ 178.37(i) *Hydrostatic test.*

* * *

- (1) Applies except that water jacket method only is authorized.
- (2) * * *
- (3) * * *
- (4) Each cylinder must be tested to at least $3/2$ times marked service pressure.

§ 178.37(j) *Toughness and ductility tests.*

- (1) Flattening Test: Between knife edges, wedge-shaped, 60-degree angle, rounded to 2-inch radius; test one cylinder taken at random out of each lot of 200 or less cylinders.

Longitudinal axis of the cylinder must be at approximately a 90-degree angle to the knife edges.

- (2) Impact Tests: For each lot of 200 or fewer cylinders, three subsize Charpy V-notch specimens must be taken from the lower sidewall of one heat treated test cylinder at approximately evenly spaced (120 degree) intervals and tested at -50 °C in accordance with ASTM E-23.
- (3) Flawed Burst Test: For each lot of 200 or less cylinders, one cylinder must be flawed and cycle tested as described in § 178.37(d)(3)(iv) of this special permit.
- (4) Hardness examination. A hardness measurement must be performed on the cylindrical section of each cylinder after heat treatment.

§ 178.37(k) *Physical test and magnetic particle examination.*

* * *

(1) * * *

(2) * * *

(i) Specimens must be: Gauge length 8 inches with width not over 1 1/2 inches; or gauge length 2 inches with width not over 1 1/2 inches. The specimen, exclusive of grip ends, must not be flattened. * * *

(ii), (iii) and (iv) * * *

(3) * * *

(4) Magnetic particle examination: All cylinders must be inspected by the wet magnetic particle method in accordance with ASTM E-709-85 before closing in and after heat treatment to detect the presence of quench cracks or other discontinuities.

§ 178.37(l) *Acceptable results of production tests and inspections.*

(1) Impact tests. The Charpy V-notch impact properties for the three specimens for each lot of cylinders must not be less than the values shown below at -50 °C:

<u>Size (mm)</u>	<u>Average value for acceptance 3 specimens</u>	<u>Minimum value 1 specimen only of the three</u>	<u>Lateral expansion inches</u>	<u>Percent fibrous fracture</u>
10x4	12.0 ft.lbs.	10.0 ft.lbs	0.012	50

(2) Flattening Test. Flattening required without cracking to eight times the wall thickness of the tested cylinder. Continue flattening until cracking occurs. Maximum degree of flattening attained without cracking and knife clearance must be entered on the inspector's report.

(3) Flawed Burst Test. The failure must originate in the cylinder sidewall and be by leakage before burst.

(4) Mechanical Tests.

(i) Tensile strength must not exceed 165,000 psi.

(ii) Elongation must be at least 16 percent for gauge length of 2 inches with width not over 1 1/2 inches.

(5) Magnetic Particle Inspection. Any cylinder found to have a quenching crack must be rejected and may not be requalified.

(6) Hardness Measurement: The tensile strength equivalent of the hardness number obtained may not be more than 165,000 psi; (HRC 37(Brinell 342)). When the results of a hardness test exceed the maximum permitted, two or more retests may be made; however, the hardness number obtained in each retest may not exceed the maximum permitted.

§ 178.37(m) *Leakage test*.

Leakage test is not required.

§ 178.37(n) *Rejected cylinders from production testing*.

(1) Reheat treatment of cylinders rejected by the impact test is authorized; subsequent thereto, acceptable cylinders must pass all prescribed tests.

(2) Reheat treatment of cylinders rejected by the flattening test is authorized; subsequent thereto, acceptable cylinders must pass all prescribed tests.

(3) Reheat treatment of cylinders rejected by the flawed burst test is authorized; subsequent thereto, acceptable cylinders must pass all prescribed tests.

(4) Reheat treatment of cylinders rejected by the mechanical properties test is authorized; subsequent thereto, acceptable cylinders must pass all prescribed tests.

(5) Cylinders rejected by the magnetic particle examination (wherein the defects are not quench cracks) may be reheat treated; subsequent thereto, acceptable cylinders must pass all prescribed tests.

(6) Reheat treatment of cylinders rejected by the hardness measurement is authorized; subsequent thereto, acceptable cylinders must pass all prescribed tests.

b. TESTING: Each cylinder must be requalified for use every five years in accordance with § 180.209 as prescribed for DOT specification 3AA cylinders except that the minimum retest pressure must be 5,250 psig as marked on each cylinder. Cylinders requalified after having been subjected to action of fire must be reported in writing to the OHMS prior to being placed in service.

8. SPECIAL PROVISIONS:

a. The manufacturer of the cylinder covered by this special permit must retain the test reports required by this special permit indefinitely as long as the cylinders are authorized for use.

b. A copy of the inspector's report for each of the first three lots produced must be submitted to the OHMS prior to shipment.

c. A person who is not a holder of this special permit, but receives a package covered by this special permit, may use the packaging authorized by this special permit for the transportation of hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

d. Each packaging manufactured under the authority of this special permit must be marked with a registration symbol designated by the OHMS for a specific manufacturing facility.

e. A current copy of this special permit must be maintained at each manufacturing facility at which this packaging is manufactured and must be made available to a DOT representative upon request.

f. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

g. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

h. These cylinders may not be used for carriage of any gas that would cause hydrogen embrittlement of the steel.

i. Filling limits specified in § 173.302a(b) are not authorized. Under no circumstance are these cylinders to be filled to a pressure exceeding the marked service pressure at 70 °F.

- j. Transportation of oxygen aboard aircraft is only authorized when in accordance with § 175.501.
- k. Before production of cylinders under this special permit may commence, the manufacturer and independent inspection agency must secure approvals in accordance with the provisions of §§ 107.803(b) and 107.803(c), respectively.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle and cargo-only aircraft (see the restriction in paragraph 8.j.) above).
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each aircraft used to transport packages covered by this special permit. The shipper must furnish a copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)—“The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR § 171.15 - Immediate notice of certain hazardous materials incidents, and

171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: TG